SciLogic: From Scientific Papers to Scientific Reasoning

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Abstract

Scientific publications communicate results through narrative prose, figures, and tables, but the structure of the argument behind their conclusions remains largely implicit: which observations are offered as evidence, which assumptions an inference depends on, and which alternative explanations remain open. Making this structure explicit is particularly difficult in empirical disciplines such as materials science and chemistry, where conclusions rest on heterogeneous and defeasible evidence rather than formal proof.

In this perspective we propose *SciLogic*, a lightweight, implementation-oriented profile for representing the published justification of a single empirical paper, and report first experience with a reference implementation. SciLogic combines established argumentation concepts, in particular from Micropublications, with three emphases: relations can record an explicit inference rule (warrant) together with its basis, quantitative evidence is kept in a form whose derived checks can be recomputed, and the annotator's assessment of each claim is stored separately from the reconstructed argument. We describe the SciLogic specification and its open reference implementation `claim-chain`, and illustrate three applications. For peer review, a worked example from computational materials science shows how an encoding makes candidate issues explicit, such as a conclusion that rests on an untested hypothesis or on an empirical rule of thumb. For finding new research directions, 62 open items collected from the encodings of ten recent preprints were matched against one research interest to propose five candidate studies. For a collaborative research project, the 62 publications of a research center were encoded and their key results summarized by research theme, together with cross-paper connections and tensions that no single paper states.

1 Introduction

Scientific publications are the principal units through which results are communicated, evaluated, and archived, but the logical structure behind their narrative is rarely explicit. A reader must reconstruct which observations serve as evidence for a hypothesis, which assumptions an inference depends on, and which alternative explanations remain open.

Over the last decade, the FAIR principles have made large parts of the scientific process findable, accessible, interoperable, and reusable [1, 2, 3]. Two components carry this effort. Ontologies and knowledge graphs provide the semantics of the data, defining what a material, a measurement, or a computed property is [4]. Scientific workflows provide its provenance, recording how a result was produced from inputs, methods, parameters, and software [5, 6], and recent work connects the two [7, 8]. Neither, however, records the inferential step through which a result becomes evidence for a conclusion. A workflow may reproducibly compute a free-energy difference, yet the claim that this difference explains an observed phase transformation is a separate inference. We therefore regard the logical chain of evidence, hypothesis, and conclusion not as a standalone representation but as the third component of reproducible science: only the provenance of the workflow, the data with its ontology, and the logical chain together cover the full scientific process.

The need for such a logical layer has become pressing. OpenAI recently released several hundred machine-generated mathematical manuscripts addressing open problems, many accompanied by formal proofs in Lean [9, 10]. Mathematics could be accelerated in this way because its reasoning has a formal representation in which every step can be checked. Language models are equally being applied across materials science and chemistry [11, 12], but there conclusions rest on heterogeneous and defeasible evidence (finite-precision measurements, approximate simulations, empirical correlations) rather than proof, and no comparable representation exists. Without one, a model that retrieves a conclusion cannot show why it is justified, and generated claims cannot be checked at scale.

Representing scientific arguments is not itself new. Argumentation models such as ScholOnto, SALT, SWAN, Micropublications, and the Argument Interchange Format represent claims, evidence, and challenges

[13, 14, 15, 16, 17, 18]. Nanopublications, SEPIO, the Evidence and Conclusion Ontology, and PROV-O capture assertions, evidence, and provenance [19, 20, 21, 22, 23, 24]. Argument mining and claim verification extract and test such structures from text [25, 26, 27, 28, 29, 30, 31, 32, 33], and scholarly knowledge graphs link them across publications [34, 35, 36, 37]. What remains open is a representation lightweight enough for routine use in empirical disciplines, grounded in the quantitative results it relies on, and reconstructable reproducibly by humans and language models alike.

We propose *SciLogic* for this purpose: a lightweight, implementation-oriented profile that combines established argumentation concepts with recomputable quantitative checks and a separately attributable assessment. SciLogic represents the *published justification* of a single empirical paper, as reconstructed by an annotator, as typed statements and typed relations that can record explicit inference rules. It reuses established concepts wherever possible and does not claim to capture how the authors actually reached their conclusions. Our contributions are:

1. the SciLogic specification, including vocabularies of statement and relation types (Section 2);
2. the open reference implementation `claim-chain` with an LLM-based converter from PDF, illustrated with a worked example from computational materials science (Sections 3 and 4.1);
3. three applications: supporting peer review, identifying open research questions across ten recent preprints, and summarizing the 62 publications of a collaborative research center (Section 4).

Whether independent annotators reconstruct the same arguments has not yet been tested, and links between the encodings of different papers are not yet represented.

2 The SciLogic specification

The specification is written as a LinkML schema [38] from which a JSON Schema and RDF exports are generated [39]. An encoding, called an *argument chain*, describes one publication and contains statements, relations between statements, the entities they refer to (materials, techniques, quantities, methods), the cited works with what each is cited for, and a paper-level assessment.

2.1 Five layers

Propositions, their discourse roles, and the source objects that carry evidence must not be conflated. SciLogic separates five layers:

1. **Discourse unit.** Each statement contains one self-contained discourse unit, written so that it can be understood on its own. For most types this is a proposition; for `QUESTION` it is a question. The text is the annotator's paraphrase; the `hedged` flag records whether the authors themselves hedge. Because no verbatim source passage is stored yet, the flag can only be checked against the paper via the locator.
2. **Discourse role.** The statement type (Table 1) is the role the proposition plays *in this paper*. For claims, the role is refined by a claim role (main conclusion, intermediate step, quantitative support, mechanistic interpretation, methodological claim, or counter-claim), a modality (descriptive, comparative, correlational, causal, generalization, conceptual), and the authors' certainty (demonstrated, predicted, proposed, speculative). The certainty categories are a pragmatic choice that mixes epistemic strength (demonstrated, speculative) with the mode of derivation (predicted, i.e. computed rather than measured).
3. **Evidence and source artifact.** Evidence is not a statement type but an attribute of a statement: its kind (citation, argument, qualitative data, quantitative data, or none), a locator pointing to the figure, table, equation, or section, the values read from it, and derived checks.
4. **Argument relation.** Typed, directed relations connect statements (Table 2). Each relation records who asserts it, how and how strongly the subject supports the object, the uncertainty of the link, and optionally the warrant behind the step, that is, the rule that licenses the inference [40].
5. **Assessment.** The annotator's verdict on each claim and hypothesis, with rationale, missing evidence, and proof obligations (Section 2.3).

Hypothesis, claim, and conclusion are not separate kinds of proposition: HYPOTHESIS marks a proposition the paper sets out to establish, and a conclusion is a CLAIM with the role *main conclusion*. When the main conclusion confirms the hypothesis, both are encoded as separate statements, possibly worded alike, and the claim SUPPORTS the hypothesis. Because each encoding covers one paper, the same proposition can be a hypothesis in one paper and an assumption or background statement in another. A figure or dataset is a source artifact referenced by a locator; it never asserts anything itself, and what the paper reads from it is recorded as an OBSERVATION. Finally, relations distinguish links stated or implied by the authors (AUTHORS) from links an annotator adds from the paper's own content (ANNOTATOR) or from outside knowledge (ANNOTATOR_BACKGROUND), and a warrant whose basis is UNSTATED marks an inference rule the paper uses without saying so. Reconstructed links and assumptions can therefore be marked as such. Links the authors state explicitly and links they only imply are not yet distinguished.

Statement type	Definition in SciLogic
BACKGROUND	Established fact taken from the literature.
GAP	Problem or gap the work addresses.
HYPOTHESIS	Proposition the paper sets out to establish (the central thesis or a sub-hypothesis).
QUESTION	Research or bridging question, including open questions.
DEFINITION	New term or concept introduced.
ASSUMPTION	Premise the argument relies on but does not test in this paper; either accepted in the community or a choice of the authors, optionally with the open question it leaves.
METHOD	Methodological choice or procedure.
DATA	Description of datasets or samples used.
OBSERVATION	Result read from this paper's own data.
CLAIM	Conclusion drawn by the authors.
LIMITATION	Limitation or inconsistency acknowledged by the authors.
OUTLOOK	Speculation or future work.

Table 1: Statement types of SciLogic. The type is the role a proposition plays in the encoded paper, not an intrinsic property of the proposition.

2.2 Relations

Table 2 lists the ten relation types. Relations are stored as directed assertions from subject to object and are read as "subject *predicate* object". Inverse and transitive edges are not generated: in particular, support is *not* propagated along paths. Some predicates nevertheless have symmetric meaning: CONTRADICTS expresses incompatibility within the same scope, so a single stored edge also implies the converse. A graph that contains a claim and a contradicting claim is valid, since it may faithfully represent a disagreement within the paper. How uncertain links should propagate to the claims that depend on them is an open design question. Endpoints are normally statements; the object of ANALOG_OF may also be an entity, such as a reference material. The validator checks only that subject and object exist as statements or entities and does not yet restrict their types, so the endpoint column lists typical rather than enforced endpoints.

Figure 1 shows the relation types in a schematic example. It is deliberately simple, but it illustrates the distinctions that matter: support by different kinds of evidence, an explicit warrant for the step from simulated migration barriers to conductivity, a scope restriction, and the difference between logical opposition and empirical counter-evidence.

Two relations need further comment. WEAKENS attacks the credibility of a conclusion. In structured argumentation this is a form of rebuttal, whereas an *undercutter* attacks the inference from premises to conclusion [41]. Undercutting in that sense requires inferences that can be addressed as objects, which SciLogic does not yet provide; the relation was therefore renamed from UNDERCUTS to WEAKENS, and older files are converted automatically. SUPPORTS, CONDITIONALLY_SUPPORTS, and DEPENDS_ON differ in the status of the subject and in how it is used. In the worked example (Section 4.1), mechanical stability (C1.3) SUPPORTS the metastability claim C1, because the paper establishes it; the untested kinetic barrier (H1) CONDITIONALLY_SUPPORTS C1, because C1 would follow if H1 were shown; and in Figure 1 the simulation O2 DEPENDS_ON the assumption A2, because it uses A2 as an input rather than offering it as evidence. CONDITIONALLY_SUPPORTS thus combines an inferential link with the unestablished status of its premise;

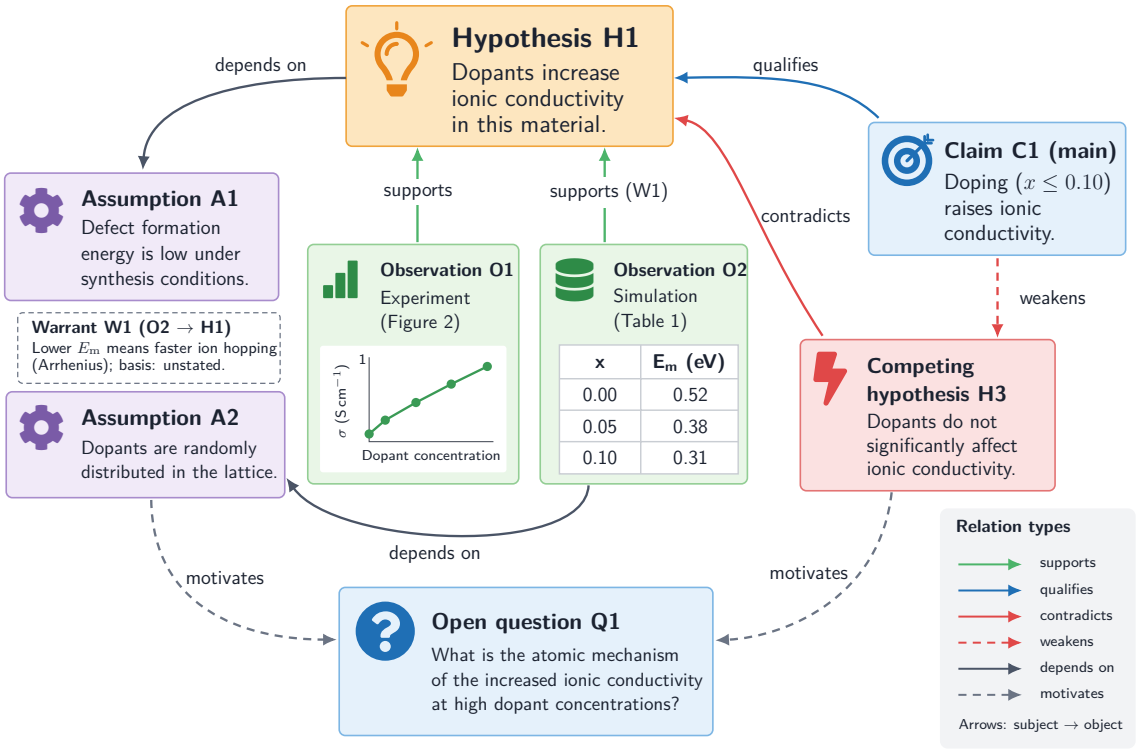


Figure 1: Schematic SciLogic graph for a hypothetical doping study. All statements and numerical values are illustrative and not taken from a publication; a real encoding is discussed in Section 4.1. Arrows point from subject to object. Both observations support hypothesis H1. The step from the simulated migration barriers (O2) to conductivity requires warrant W1, which the hypothetical paper leaves unstated. The main claim C1 qualifies H1 by restricting it to the tested range. H3 contradicts H1, since both cannot hold in the same scope, whereas C1 weakens H3: it is empirical counter-evidence that lowers the credibility of H3 without refuting it outside the tested range. The simulation O2 depends on assumption A2 of randomly distributed dopants. Because this assumption is likely to fail at high dopant concentrations, A2 motivates the open question Q1, as does the competing hypothesis H3.

Predicate	Definition (subject → object)	Typical endpoints
SUPPORTS	Gives evidence or a reason that makes the object more credible.	observation, claim → claim, hypothesis
CONDITIONALLY_SUPPORTS	Would support the object if the subject were shown; used for untested hypotheses the argument leans on.	hypothesis → claim
QUALIFIES	Restricts the scope or strength of the object without refuting it.	claim → claim, hypothesis
CONTRADICTS	Cannot hold together with the object within the same scope (symmetric in meaning).	claim, hypothesis → claim, hypothesis
WEAKENS	Counter-evidence: lowers the credibility of the object without contradicting it.	observation, claim → claim, hypothesis
ALTERNATIVE_EXPLANATION_FOR	Offers another cause for an observation or claim.	hypothesis, claim → observation, claim
MOTIVATES	Gives the reason to pose or investigate the object.	any → question, hypothesis
ANSWERS	Answers the question posed by the object, fully or in part.	claim, observation → question
DEPENDS_ON	Uses the object, e.g. a method uses a definition or a simulation an assumption.	method, observation, hypothesis → assumption, definition
ANALOG_OF	Is defined by analogy to the object.	statement → statement, entity

Table 2: Relation types of SciLogic. The definitions are those documented in the schema [39].

recording that status on the statement and using SUPPORTS throughout is an alternative we consider for a future version. A warrant on a binary relation explains one step but does not represent a multi-premise inference; Section 5 returns to this limitation.

2.3 Assessment as a separate layer

SciLogic does not decide whether a claim is true. The assessment layer records a narrower judgment: *does the data presented in the paper support the claim?* Each claim and hypothesis receives one of five verdicts with a rationale, the missing evidence, and *proof obligations*. The annotation instructions define the verdicts operationally: YES, the presented data support the claim as worded; PARTIAL, part of the claim is shown (for example a trend) and part is not (for example the mechanism, the generality, or a comparative "better"); WEAK, all support is indirect, thin, or uncontrolled; INCONCLUSIVE, no data in the paper decide the claim, for example when it is argued by analogy or a mechanism is proposed but not tested; and NO, the paper's own data or statements contradict it. NO thus means contradicted, not merely unsupported, and INCONCLUSIVE means undecided by the presented data. A proof obligation states what would have to be shown for the claim to hold, with its status, priority, and the data that would discharge it; an open obligation is a candidate request a reviewer might make. Cited references are not looked up; they are assumed to provide what they are cited for. The verdict is therefore conditional on the cited support being represented correctly, and it evaluates the internal logic of the paper, not the state of the field. The generated reports do not yet state this condition explicitly.

The implementation also derives draft scores from the verdicts: a logic score (the mean of YES $= 1$, PARTIAL $= 0.5$, WEAK $= 0.375$, INCONCLUSIVE $= 0.25$, NO $= 0$ over all claims and hypotheses), a variant weighted by claim role, and the fraction of discharged proof obligations. These weights are arbitrary and unvalidated; in particular, ranking INCONCLUSIVE below PARTIAL has no measurement basis. The mean and the fraction of discharged obligations also change when an annotator splits one claim into several. The score is an experimental index for comparing encodings of the same paper. It is neither a probability that the paper is correct nor a measure of its quality, and we report verdict counts alongside it. Because the assessment is an annotator's judgment, it must remain separable from the reconstructed argument. Two annotators may agree on the argument and disagree on its assessment, and both outcomes are informative. The current validator, however, requires an assessment for every claim and hypothesis, so a reconstruction cannot yet be validated on its own; a separate validation profile for unassessed reconstructions would make

the conceptual separation operational. Whether assessments should be stored in a file separate from the encoding is an open question.

2.4 Relation to the closest models

SciLogic is not a new ontology but a lightweight profile that combines established argumentation concepts with quantitative checks and a separately attributable assessment. Table 3 compares it with three of the closest models. The entries summarize the primary descriptions cited; a feature listed as not described may still be expressible through extensions of a model.

Aspect	Micropublications [16]	SEPIO [21]	AIF [17]	SciLogic
Inference step	Support and challenge relations between claims, statements, and evidence	Evidence lines linking evidence to an assertion, with direction and strength	Inference nodes typed by argument scheme	Typed binary relations with optional warrant and its basis
Joint premises	Not described	An evidence line groups several evidence items	Several premises per inference node	Not yet represented
Evidence source	Data, methods, and materials as supporting elements	Evidence items with provenance of agents, methods, and tools	Not a focus	Locator to figure, table, equation, or section
Quantitative values	Not described	Not described	Not described	Values with units; checks recomputed by the software
Attribution of links	Attribution of statements and annotations	Provenance of assertions and evidence lines	Not a focus	Authors, annotator, or annotator with outside knowledge, per relation
Assessment	Not described	Confidence and strength of evidence lines	Preferences between arguments	Separate verdict layer with proof obligations
Intended workflow	Annotation of biomedical communications	Integration of curated evidence	Exchange between argumentation tools	LLM-assisted encoding and review of single papers

Table 3: SciLogic compared with three closely related models. Entries for the other models summarize the cited primary descriptions and do not exclude extensions.

3 Reference implementation

The open-source package `claim-chain` [39] implements SciLogic. It consists of:

- the LinkML schema, with a JSON Schema generated from it that can be used for structured LLM output, and RDF export;
- a renderer that turns an encoding into a readable argument chain, a Markdown report with verdicts, limitations, and missing evidence, argument maps, and the draft scores, and that recomputes all derived checks;
- a validator that checks an encoding against the schema and for referential consistency, for example that identifiers are unique, that relations refer to defined statements, and that every claim and hypothesis has an assessment;
- a command-line tool and Python API, `claimchain`, that converts a PDF into an encoding with an LLM (Claude). The model reads the PDF including its figures. The returned encoding is validated, and any problems are sent back to the model for up to two rounds of correction;

- annotation instructions and a review checklist, used both by the converter and for LLM-assisted manual annotation;

The converter does not compute scores or run the derived checks; the renderer is run on its output for that. Its output has not yet been evaluated against hand-checked encodings. The repository contains two worked examples: a hand-encoded recent paper with 64 statements, and the example discussed next. It also contains LLM-drafted encodings of ten recent materials-science preprints from arXiv with 27 to 36 statements each.

4 Applications

We see three uses of SciLogic, each serving a different reader: a reviewer who has to judge whether a paper's conclusions follow from its evidence (Section 4.1), a researcher looking for the next project in the open questions of a field (Section 4.2), and a research consortium summarizing what its body of work has established, for example for a report or a follow-up proposal (Section 4.3). The first works on a single paper, the other two on collections of papers.

4.1 Supporting peer review

A reviewer has to reconstruct the logical structure of a paper before judging it. A SciLogic encoding makes one such reconstruction explicit and open to inspection. As a documented case we use the encoding of Zhuang, Chen, and Carter (2017), who predict a $D0_{19}$-ordered Mg_3Al intermetallic compound with orbital-free and Kohn–Sham density functional theory and characterize its stability and ductility [42].

Id	Statement (role)	Relation and warrant	Verdict
C1	$D0_{19}$ Mg_3Al is at least metastable (main conclusion).	Supported by C1.1–C1.3; qualified by C1.4.	`PARTIAL`
C1.3	$D0_{19}$ Mg_3Al is mechanically stable (intermediate).	Strongly supports C1; warrant: Born criteria (basis: citation), recomputed.	`YES`
C1.4	$D0_{19}$ Mg_3Al is thermodynamically unstable against decomposition into $Mg_{17}Al_{12}$ and Mg (counter-claim).	Qualifies C1: it does not refute metastability, but makes it depend on a kinetic barrier.	`YES`
H1	A large kinetic barrier may prevent this decomposition (hypothesis).	Conditionally supports C1; warrant: a locally stable phase persists if the decomposition barrier is large (basis: domain convention).	`INCONCLUSIVE`
C2	$D0_{19}$ Mg_3Al is predicted to be more ductile than hcp Mg (main conclusion).	Supported by four descriptors C2.1–C2.4, each with its own warrant.	`PARTIAL`
C2.1	Larger Pugh ratio B/G than hcp Mg (quantitative support).	Supports C2; warrant: Pugh criterion (basis: empirical correlation).	`YES`
C2.4	Low anti-phase boundary energy compared with Ni_3Al (quantitative support).	Supports C2, but weakens the claimed ordering $D0_{19} > L1_2$ within C2 (link added by the annotator).	`YES`

Table 4: Selected statements of the worked example (Zhuang et al. 2017). The full encoding, report, and argument maps are in the repository [39]. Verdicts were assigned with LLM assistance during conversion to the current schema.

The encoding contains 21 statements (15 claims and hypotheses, two assumptions, and four open questions), 20 relations, and 22 derived checks, all of which reproduce from the recorded values. The paper has two main conclusions: $D0_{19}$ Mg_3Al is at least metastable (C1), and it is predicted to be more ductile than hcp Mg (C2). In the encoding, metastability means local stability: the structure is dynamically and mechanically stable and energetically competitive with other Mg_3Al structures, while the paper itself finds it thermodynamically unstable against decomposition (C1.4). Table 4 shows selected statements. Of the 15 claims and hypotheses, 9 are assessed `YES`, 3 `PARTIAL`, and 3 `INCONCLUSIVE`; 4 of the 8 proof obligations are discharged. Thirteen of the 20 relations are attributed to the authors and seven to the annotator. Six of the twelve support relations carry an explicit warrant (two with the basis domain convention, two

author argument, one citation, one empirical correlation), and six statements carry a warrant for their own evidence; none of these warrants is marked as unstated by the paper.

Two of the derived checks illustrate the quantitative layer. For mechanical stability (C1.3), the encoding records the elastic constants $C_{11} = 73$, $C_{12} = 33$, $C_{13} = 30$, and $C_{33} = 73$ GPa from the paper's Table 1 and evaluates the Born criterion $2C_{13}^2 < C_{33}(C_{11} + C_{12})$, which gives $1800 < 7738\,\mathrm{GPa}^2$ and is satisfied. For ductility (C2.1), it records the Pugh ratios $B/G = 2.65$ for $D0_{19}$ Mg_3Al and 1.76 for hcp Mg and computes an increase of 50.6 %. That all 22 checks recompute shows that the recorded calculations are internally consistent. It does not validate the transcription of the values, the applicability of the criteria, or the conclusions drawn from them.

In this reconstruction, the annotator encodes the paper's own counter-claim (C1.4) as a qualification of C1 rather than a contradiction. The first main conclusion then depends on a hypothesis (H1) that the paper does not test, and the annotator records the corresponding open proof obligation, a transformation path and its barrier, as a candidate issue for review. The four ductility descriptors are encoded as separate support relations, each with its own warrant; they are separate lines of argument in the paper, not necessarily independent evidence. One of these warrants, the Pugh criterion, is recorded with the basis *empirical correlation*, which flags it for a reviewer's attention. Finally, one result (C2.4) both supports and weakens C2, because C2 bundles a comparison with hcp Mg and an ordering of two Mg_3Al structures. The encoding thus points to a compound claim that should be split; we have kept it unsplit here to show how the structure exposes such cases. Whether a domain expert would identify the same issues has not been checked.

4.2 Identifying open research questions

A researcher looking for the next project typically gathers recent papers in the field, notes what they leave open, and selects the questions that match their own expertise. SciLogic makes the first two steps mechanical. Open items are already explicit in an encoding, but they have different origins: limitations and outlook statements are stated by the authors, the open question of an assumption is formulated by the annotator, and open proof obligations are proposed by the annotator. Because every item keeps the identifier of the statement it comes from, research ideas derived from them remain traceable to the papers.

Proposed study	Open items addressed	Source papers
S1 Transferability of grain-boundary shear strengths across interatomic models	Dependence on the interatomic potential, metastable starting structures, compact supercells, reproducibility of generated code, missing comparison with a conventional workflow (5)	2610.28629 [43]
S2 Atomistically parameterized interfaces for phase-field models of Zr oxidation	Assumed eigenstrain and orientation relationship, averaged boundary energies, chosen nucleation parameters, 2D without plasticity, untested link to cracking (5)	2610.28493 [44]
S3 Size-dependent phase stability in nanocrystalline thin-film libraries	Retention of non-equilibrium phases in films with nanometre grains; transfer of film-based rankings to bulk (4)	2610.28232 [45], 2610.27355 [46]
S4 Stability and storage mechanism of Co-doped Janus HfSSe	Positive dopant formation energy, unshown synthesizability, ionic storage versus metal plating in outer layers (3)	2610.28152 [47]
S5 Simulation-derived structural bases for differentiable powder diffraction	Motions outside the neutron-derived basis, abrupt transitions excluded by the shared model, supply of simulated kinetics (3)	2610.28564 [48]

Table 5: Research studies derived from the open questions of ten recent arXiv preprints for a research interest in computational materials science. Numbers in parentheses give the number of open items addressed. Matching and study design are a single LLM pass (Claude) over the encodings and have not been reviewed by a domain expert. The catalogue and the studies are in the repository [39].

We applied this to the ten arXiv preprints that serve as test papers in the repository. They were taken from the new submissions to cond-mat.mtrl-sci listed on 21 October 2026, chosen to cover experimental, computational, high-throughput, characterization-method, and AI-tool papers, and encoded from the PDFs of their first versions by an LLM (Claude) on the same day [49, 50, 46, 51, 52, 47, 45, 44, 48, 43]. A short script collects 62 open items from their encodings without deduplication: 29 stated by the authors

(19 limitations and 10 outlook statements), 20 questions formulated by the annotator for assumptions, and 13 open proof obligations proposed by the annotator. In a single pass, the LLM then classified each item against one research interest, computational materials science with atomistic simulation, reproducible workflows, and coupling to mesoscale models and experiment: 17 items can in its judgment be addressed with the methods of this research interest alone, 18 only in combination with experiments, and 27 lie outside it. From these 35 items it derived five studies (Table 5), which address 20 of them; the remaining 15 were not developed further.

Most studies address the open questions of a single computational paper, as expected for a computational research interest. Study S3 is different: it combines open questions from two unrelated experimental papers, a combinatorial screening of fusion materials on co-sputtered films [45] and an on-chip annealing method for films with 3–7 nm grains [46], which both leave open whether nanocrystalline films retain phases that are not stable in bulk. Neither paper states this shared question; here it emerged from comparing the open items of several papers. We have not tested whether a direct reading of the papers, or an LLM given their full texts, would identify it as readily. The example also shows the limits of the current test: the encodings are LLM drafts, the matching was done without re-reading the papers, and the value of the proposed studies can only be judged by researchers in the field.

4.3 Summarizing a collaborative research project

The third use starts from a body of work rather than from a field. When a consortium reports on a funding period or prepares a follow-up proposal, it has to state what its publications have established, which results rest on assumptions that were never tested, and which proof obligations remain open. We applied SciLogic to the complete publication list of the Collaborative Research Center CRC 1625 on compositionally complex solid solution surfaces for electrocatalysis: 62 publications from 2024 to 2026 (57 journal articles and 5 preprints), covering combinatorial thin-film synthesis, electrochemistry, atomistic simulation, and research data management. An LLM (Claude) encoded each publication from its PDF, following the annotation guide of the repository but without the automated extraction pipeline of `claim-chain`; figures were judged from their captions and the surrounding text. All 62 encodings pass schema validation, and all of their derived checks evaluate.

Research theme	Key results (verdict of the main conclusion)	Publications
Synthesis and atomic-scale characterization platforms	Thermal SiO_2 barrier extends Si-microtip platforms to $\geq$600 °C (`YES`); kinetic, history-dependent onset of a Pt-rich phase down to 300 °C (`YES`); epitaxial films for correlative structure–activity mapping (`PARTIAL`)	[53, 54, 55, 56]
High-throughput and autonomous discovery	Activity tunable by composition across four reactions (`YES`); non-metal-hybridized Pt alloys for oxygen reduction (`YES`); activity trend of Au–Ir–Rh predicted from 15 % of the measurement areas (`PARTIAL`)	[57, 58, 59]
Descriptors of the electrochemical interface	Coverage-corrected adsorption energy subsumes the work function (`YES`); hydrogen coverage and H–H repulsion govern hydrogen evolution on alloys (`PARTIAL`); submicron measurement of the potential of zero charge (`PARTIAL`)	[60, 61, 62, 63]
Surface composition and restructuring	Ag enrichment and Pt depletion at Ag–Au–Pd–Pt surfaces (`PARTIAL`); activation of the Cantor alloy by a NiFe-rich oxyhydroxide layer (`PARTIAL`)	[64, 65]
Atomistic simulation methods	Machine-learning potentials with near-DFT accuracy (`YES`, review); thermopotentiostat for electrified interfaces (`PARTIAL`)	[66, 67]
Research data and data-driven methods	Extensible data types in the data management system (`YES`); ontology and knowledge graph (`PARTIAL`); active learning with sevenfold fewer measurements (`PARTIAL`); text-derived composition embeddings for screening (`WEAK`, `NO`)	[68, 69, 70, 71, 72]

Table 6: Key results of the CRC 1625 publications, grouped by research theme, with the verdict of the corresponding main conclusion. The encodings are LLM drafts (Claude) and have not been reviewed by the authors or by domain experts. The 62 encodings are in the repository [39].

The 62 encodings contain 955 statements and 1,109 relations, of which 952 are attributed to the authors and 157 to the annotator. Of the 127 main conclusions stated in results and discussion sections, 9 are assessed `YES`, 85 `PARTIAL`, 24 `WEAK`, 5 `INCONCLUSIVE`, and 4 `NO`. Of 199 derived checks, 17 record a reported number that does not follow from the paper's own values, for example a stress-to-strain conversion that is off by a factor of four. Of 100 proof obligations, 20 are discharged, 22 partially discharged, and 58 open. The most frequent concerns are internal inconsistencies (50), missing quantification (47), untested assertions (30), and confounded comparisons (28).

Table 6 groups the key results into six research themes. The nine main conclusions assessed `YES` indicate which results the presented evidence supports most firmly: a thermally grown SiO_2 barrier that extends Si-microtip platforms for atom probe tomography from 300 to at least 600 °C [53, 54]; the finding that the onset temperature of Pt-rich phase formation in $Ag_{24}Au_{20}Pd_{50}Pt_6$ is a kinetic observation boundary rather than a phase boundary [55]; composition-tunable activity across four reactions and several alloy systems [57]; Pt-based alloys hybridized with non-metals that exceed benchmark catalysts in oxygen-reduction activity and durability [58]; the result that a coverage-corrected hydrogen adsorption energy captures what the work function adds as a descriptor [60]; the extensible type system of the project's data management system [68]; and, as a literature synthesis, the near-DFT accuracy of machine-learning interatomic potentials for compositionally complex alloys [66]. A `YES` verdict means that the paper's own data support the claim as worded, not that the result is the most important one; judging impact would require expert assessment or citation data.

Reading the encodings side by side also shows connections that no single paper states. Three independent studies, a computational one [61], a combined experimental and computational descriptor study [60], and an in-situ spectroscopy study [62], converge on hydrogen coverage as the quantity that determines hydrogen evolution on these surfaces; together they suggest the adsorption energy at operating coverage as a shared descriptor for the next funding period. The Ag–Au–Pd–Pt system serves as a common model system in five publications and links them into a methodological sequence, from the failure of the native oxide at 400 °C [53], through the thermal oxide barrier [54], to the reconstruction of a hidden transformation pathway [55]. The same comparison also exposes tensions: the composition of the Pt-rich phase at 600 °C is reported as about 22 and about 40 at. % Pt in two papers on the same alloy [54, 55]; the descriptor study does not model the surface segregation that a companion paper measures for the same alloy family [60, 64]; and two papers on the data management system report FAIR self-assessment scores of 77 and 82 % for different instances [69, 73]. Recurring open obligations, such as normalizing activities to the electrochemically active area rather than the geometric area, or comparing data-driven screening with a random baseline, point to work packages that would strengthen several papers at once.

The analysis has clear limits. The encodings are LLM drafts that neither the authors nor domain experts have reviewed, and their verdicts judge only whether each paper's own evidence supports its claims. The cross-paper connections were found by reading the encodings, not computed, because SciLogic does not yet represent links between encodings of different papers. Turning such a summary into a proposal remains the consortium's task; the encodings make the starting point explicit and traceable to individual statements.

5 Discussion and limitations

SciLogic occupies a space between formal logic, semantic knowledge representation, and scientific narrative. It does not turn empirical science into proof: a relation such as `SUPPORTS` remains an evidential judgment whose strength depends on uncertainty, conditions, model assumptions, and disciplinary conventions. Its value lies in making that judgment explicit and attributable, so that it can be inspected, disputed, and compared.

The present work has clear limitations. A warrant attached to a binary relation explains one inferential step, but the specification cannot yet represent joint premises, assumptions attached to a single inference, or attacks on an inference itself; it therefore does not capture the complete inferential structure of an argument. Its provenance stops at the level of figures, tables, and sections, and paraphrased statements are not anchored to verbatim passages. Links stated by the authors and links they only imply are not distinguished. Endpoint types of relations are not enforced, and the validator requires an assessment even when only a reconstruction is wanted. Verdicts are conditional on cited works providing what they are cited for. The scores are experimental indices without agreed weightings. The worked example was assessed with LLM assistance, and the research studies derived from open items are proposals from a single LLM pass, and the 62 encodings of the research-center publications are LLM drafts; no domain expert has yet

reviewed any of them. Finally, this perspective reports no reproducibility data. Whether experts converge on the same argument graph, and where they legitimately do not, remains the open empirical question. We consider it the most important question for the approach: if independent experts consistently disagree about the role of a statement or the strength of an inference, the disagreement itself is information about how explicit the reasoning in contemporary publications is.

6 Conclusion

SciLogic represents the published justification of an empirical paper as typed statements, typed relations with explicit warrants, quantitative evidence that can be recomputed, and an assessment kept separate from the reconstructed argument. Its specification is available as a versioned schema with an open reference implementation. A worked example illustrates how the structure makes candidate weaknesses of an argument explicit for a reviewer, and the open items collected from the encodings of ten recent preprints can be matched against a research interest to propose candidate studies, which still require evaluation by experts. The next step is the expert study that tests whether published scientific arguments can be reconstructed reproducibly enough to support inspection, and that shows where such reconstruction remains ambiguous.

Data and code availability

The SciLogic schema, the `claim-chain` reference implementation, the two worked examples, the ten LLM-drafted test-paper encodings, the 62 LLM-drafted encodings of the CRC 1625 publications, and the catalogue of open questions with the derived research studies are openly available at https://github.com/lab-mids/claim-chain [39]. This manuscript refers to commit `cb8a248`. The schema is released under CC-BY-4.0 and the software under GPL-3.0. The PDFs of the test papers are not redistributed; a script in the repository retrieves them from arXiv. The RDF namespaces (`w3id.org/claim-chain/...`) are placeholders and not yet registered.

Author contributions

Conceptualization, methodology, software, validation, visualization, writing, and supervision contributions will be specified in the final manuscript.

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